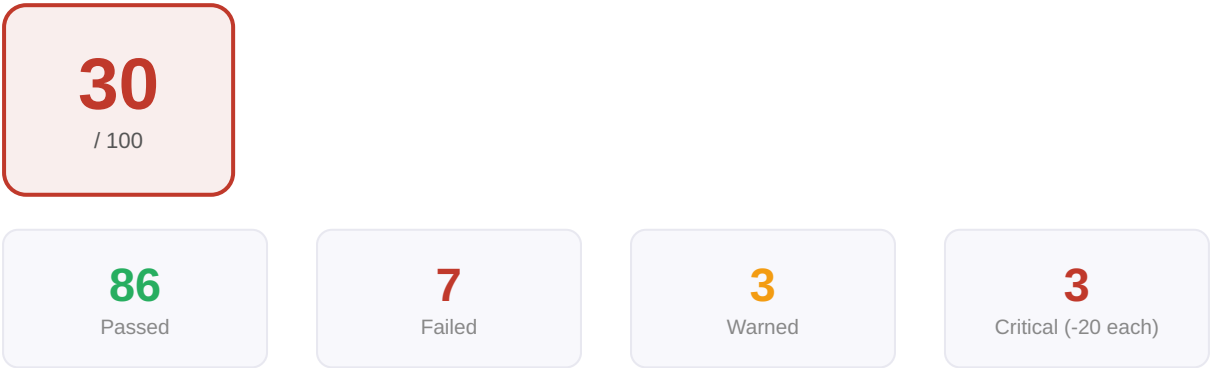


Data Contract Enforcer — Weekly Report

Generated: 20260402 | Pipeline: TenX Academy Week 7 | Contracts validated: 6 | Total checks: 96

1 • Data Health Score



Pipeline health is critically low at 30/100 (3 critical violation(s) active); the pipeline should be halted until root causes are resolved.

Formula: base = 89.6 (86/96 x 100) – 3 x 20 = 30

2 • Violations This Week

Count by Severity

Severity	Count
CRITICAL	3
HIGH	1
MEDIUM	3
WARNING	3

Top 3 Failing Checks

- [CRITICAL] LangSmith Traces - end_time > start_time: 150/150 violations. Affects 150 record(s). Impact: downstream time-series analytics and SLA calculations will produce incorrect results. Detail: 150 records violate 'end_time > start_time'.
- [CRITICAL] Week 1 – Intent Records - intent_id: 1 duplicates in 2 values. Affects 1 record(s). Impact: downstream consumers may surface incorrect data. Detail: 'intent_id' has 1 duplicate values. Duplicates indicate an idempotency violation in the pipeline.
- [CRITICAL] Week 5 – Events - recorded_at >= occurred_at: 100/100 violations. Affects 100 record(s). Impact: downstream time-series analytics and SLA calculations will produce incorrect results. Detail: 100 records violate 'recorded_at >= occurred_at'.

3 • Schema Changes Detected (Past 7 Days)

4 change(s) detected - 3 breaking, 1 compatible.

Dataset	Field	Change Type	Verdict
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				Plain Summary	Required Action
Week 3 – Extractions	review_notes	ADD_NULLABLE_COLUMN	COMPATIBLE	field 'review_notes' added (nullable) - no action needed	None. Downstream consumers can ignore the new column.
Week 3 – Extractions	doc_id	REMOVE_COLUMN	BREAKING	field 'doc_id' removed - any consumer reading this field will break	Deprecation period mandatory (minimum 2 sprints). Blast radius report required. Each affected consumer must acknowledge removal in writing (JIRA ticket or PR comment).
Week 3 – Extractions	review_notes	REMOVE_COLUMN	BREAKING	field 'review_notes' removed - any consumer reading this field will break	Deprecation period mandatory (minimum 2 sprints). Blast radius report required. Each affected consumer must acknowledge removal in writing (JIRA ticket or PR comment).
Week 3 – Extractions	doc_id	ADD_NONNULLABLE_COLUMN	BREAKING	field 'doc_id' added (NOT NULL) - producers must supply a value	Coordinate with all producers. Provide a default or migration script. Block deployment until all producers updated.

4 · AI System Risk Assessment

Overall AI Risk: LOW - Overall AI system risk: LOW. AI pipeline is operating within acceptable parameters.

❑ **Embedding Drift** [SKIPPED]
SKIPPED

❑ **Prompt Input Schema** [PASS]
0/50 records quarantined (0.0%). Prompt inputs are well-formed.

❑ **Output Schema Violations** [PASS]
Violation rate=0.0% (threshold=2.0%). Trend vs baseline: stable. Rate within acceptable bounds.

5 · Recommended Actions

#1 **CRITICAL** `src/langsmith/ingest.py`

Fix LangSmith trace ingestion to populate `end_time` correctly: update `src/langsmith/ingest.py` to set `end_time = run.end_time` (not null). All 150 traces currently have `end_time ≤ start_time`, breaking SLA dashboards.

#2 **CRITICAL** `src/week5/event_emitter.py`

Fix event timestamp population in the Week 5 event emitter: update `src/week5/event_emitter.py` so `recorded_at` is set after `occurred_at` (currently 100/100 events violate this constraint), or swap the field assignment to use UTC clock at emit time.

#3 **CRITICAL** `src/week1/intent_router.py`

Resolve duplicate `intent_id` in Week 1 intent records: update `src/week1/intent_router.py` to generate a new UUID per record instead of reusing 'INT-001'. Duplicate IDs break idempotency guarantees for downstream deduplication.